D214 Data Analytics Graduate Capstone

NKM2 Presentation of Findings (Task 3) Performance Assessment

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Executive Summary

Cardiovascular disease (CVD) is the leading cause of death worldwide and continues to pose challenges for healthcare systems (CDC, 2024). Managing CVD often requires long-term treatment and costly procedures, creating a substantial financial burden, while hospitals and clinics face increased demand for care due to its widespread prevalence, which can overwhelm healthcare providers and limit the availability of resources for other patients. The purpose of this study is
investigate the question "Do demographic, lifestyle, and clinical factors significantly impact the likelihood of cardiovascular disease?" The research is guided by the following hypotheses:

- H_0 (null hypothesis): None of the predictor variables are significantly associated with cardiovascular disease.
- H_1 (alternative hypothesis): At least one variable is significantly associated with an increased likelihood of cardiovascular disease.

Data Analysis Process

The analysis uses "Cardiovascular disease dataset," a merged dataset from the UCI Machine Learning repository's Heart Disease dataset and a separate Kaggle "Heart Disease" dataset. (Welkins, n.d.) In total, it contains 68,205 records and 17 variables as follows:

- id
- age
- gender
- height
- weight
- ap_hi
- ap_lo
- cholesterol
- gluc
- smoke
- alco
- active
- cardio (target variable)
- age_years
- bmi
- bp_category
- bp_category_encoded

Data preparation involves removing irrelevant columns, dropping duplicates, creating dummy variables for categorical predictors, and handling extreme BMI values with winsorization. Exploratory analysis includes descriptive statistics and visualizations of univariate and bivariate relationships, indicating that patients with cardiovascular disease generally had higher BMI and older age distributions compared to those without the disease. Similar patterns are seen for blood pressure and cholesterol, where elevated levels are more common among patients with the disease. Other lifestyle factors, such as physical inactivity, alcohol use, and smoking status, showed less consistent relationships.

Predictor significance is evaluated using p-values and multicollinearity is assessed using variance inflation factors (VIF). The dataset is split into training (80%) and test (20%) sets to ensure unbiased model evaluation. Backwise elimination is used to refine the model by removing statistically insignificant or highly correlated variables, resulting in three successive models. Model performance is then assessed using accuracy score and a confusion matrix. Cook's distance is used to detect influential outliers, and the odds ratios are calculated to interpret the strength and direction of each predictor's association with cardiovascular disease.

Key Findings

The final logistic regression model retains 11 predictors: "bmi," "smoke," "alco," "active," "gender_male," "cholesterol_above_normal," "cholesterol_well_above_normal," "glucose_well_above_normal," "bp_elevated," "bp_hypertension_stage_1," and "bp_hypertension_stage_2." The model has a pseudo R^2 of 0.1358 and an accuracy score of 70%. The model is found to have an accuracy score of 70% and the confusion matrix shows that the model accurately identifies 4,675 patients without CVD and 3,101 with CVD, but misclassifies 1,472 patients as disease-free and 2,243 as having the disease.

After an odds ratio analysis, it is found that hypertension stage 2 and well above normal cholesterol are the strongest predictors, increasing odds of disease by more than 10 fold and 3.5 fold, respectively. Being male increases odds of CVD by ~5%. Surprisingly, smoking and alcohol consumption shows odds ratio below 1, suggesting lower disease odds; however, this likely reflects confounding factors such as age.

Limitations

The dataset explains only 13.58% of the variance of the variance in CVD outcomes, leaving most variation unexplained. Missing factors such as genetics, diet, stress, and medication use could account for this gap. In addition, reliance on self-reported lifestyle measures, such as physical activity and alcohol usage, introduces potential bias. Logistic regression's assumption of linearity of the logit can also oversimply how real-world health factors work together due to non-linear patterns being missed.

Recommendations

Results suggest prioritizing early monitoring and interventions for patients with stage 2 hypertension and elevated cholesterol levels, as these groups face the greatest risk for CVD. Healthcare providers and public health programs should emphasize preventative care focused on blood pressure and cholesterol management, alongside weight control strategies.

Two avenues for research research are proposed:

1. Incorporating additional variables related to family history or dietary information, which could improve explanatory power
2. Explore advanced modeling techniques like random forests, which may better capture nonlinear relationships and improve predictive accuracy beyond logistic regression

Expected Benefits of Study

Implementing the findings of this study has the potential to improve both patient outcomes and healthcare efficiency. The model showed that patients with stage 2 hypertension were over 10 times more likely to have cardiovascular disease (CVD), and those with very high cholesterol were 3.5 times more likely. Targeting these high-risk groups for early monitoring and treatment could substantially reduce progression of CVD in thousands of patients by 30-40% (Feingold, 2025)

With a 70% accuracy rate, the model already improves upon random guessing (50%) and correctly identifies 4,675 patients without CVD and 4,124 with CVD. If integrated into clinical screening, this could mean more than 8,700 patients in this dataset alone were accurately classified, guiding care in the right direction. Even a modest improvement of 5 percentage points in accuracy would reduce misclassifications by over 500 patients, minimizing unnecessary treatments for false positives and ensuring timely intervention for false negatives.

By prioritizing interventions for hypertension, cholesterol, and weight management, healthcare providers can use these results to improve early detection, reduce misdiagnosis, and optimize allocation of medical resources. In practice, this could ease demand on hospital systems, lower costs associated with late-stage CVD treatment, and improve quality of life for at-risk patients.

Sources

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